FORECASTING FOR BUSINESS

Revised Exam Study Guide — Unit 1 & Unit 2

Simple-language explanations + 3-Mark exam answers + fully worked case study

WHAT THIS FILE CONTAINS

Unit 1 (3 learning hours): forecast types, qualitative methods, time-series components, ACF/correlogram, error metrics (MAD, MSE, MAPE, MPE), error analysis (Ljung-Box, homoscedasticity, Shapiro-Wilk).

Unit 2 (6 learning hours): Simple Moving Average, Weighted Moving Average, Double Moving Average, Simple Exponential Smoothing.

Every topic has a plain-language explanation, a real-life example, and a short '3-Mark Answer' box written the way you would write it in an exam.

One full case study — Sunrise Bakery's weekly bread sales — solved step by step with genuine, hand-checkable numbers for every method, so you can see how SMA, WMA, DMA and SES actually differ on the same data.

HOW TO USE THIS GUIDE

1. Read the plain-language idea first — don't start with the formula.
2. Look at the real-life example so the idea has a picture attached to it.
3. Study the worked numbers — every number here is computed, not invented.
4. Use the 3-Mark Answer box as your exam template: define → why it matters → how it's checked → interpret → one limitation.
5. Finish each unit with the Rapid Revision table at the end.

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UNIT 1 — Introduction to Time Series Forecasting

⏱ WHY THIS UNIT HAS 3 LEARNING HOURS

Unit 1 is foundation-building, not heavy computation — you are learning vocabulary and ways of thinking (what a forecast is, how to classify it, how to judge if a forecast was good) rather than running a long calculation.

Because it's conceptual, 3 hours is enough time to cover it well, but 'fewer hours' never means 'less content' — it means the content leans more on understanding definitions and reading charts than on multi-step arithmetic.

1.1 What is forecasting?

Forecasting means using the information you have right now — usually past data — to make an informed estimate of what will happen in the future. Businesses forecast because almost every decision (how much stock to buy, how many staff to schedule, how much cash to keep ready) has to be made before the future actually arrives.

IN PLAIN ENGLISH

Think of forecasting like checking the weather before you pack for a trip. You don't know for certain it will rain, but yesterday's clouds and today's forecast give you a reasoned guess — enough to decide whether to pack an umbrella.

REAL-LIFE EXAMPLE

A bakery owner looks at how many loaves sold each day last month to decide how much flour to order this week. The forecast is not a guarantee — a surprise rainy day could change footfall — but it is far better than guessing blindly.

The five-step forecasting process

1. Formulate the problem and collect data — decide what you are forecasting, for whom, and over what time horizon.
2. Manipulate and clean the data — check for missing values, duplicate entries, and whether old data is still relevant.
3. Build and evaluate a model — choose a method that fits the pattern in the data and compare forecast errors.
4. Implement the model — produce the forecasts you actually need, ideally testing the method on data you held back.
5. Evaluate the forecasts — compare forecasts to what really happened, study the errors, and revise if the errors show a pattern.

Data-quality rule: a clever model cannot rescue bad data. This is often summarised as "garbage in, garbage out."

✓ 3-MARK ANSWER — Q1. What is forecasting, and why can it never be guaranteed?

1. Forecasting is using current and past information to estimate a future value needed for a business decision.

2. It can never be guaranteed because the future can differ from the past — new competitors, price changes, weather, or unexpected events can all break the pattern.

3. A forecast is therefore a reasoned estimate, not a promise, and should always be reviewed against actual results.

1.2 Types of forecasts

Forecasts are classified along three independent dimensions: how far ahead they look, how broad their scope is, and what method/output they use.

Figure 1. How forecasts are classified

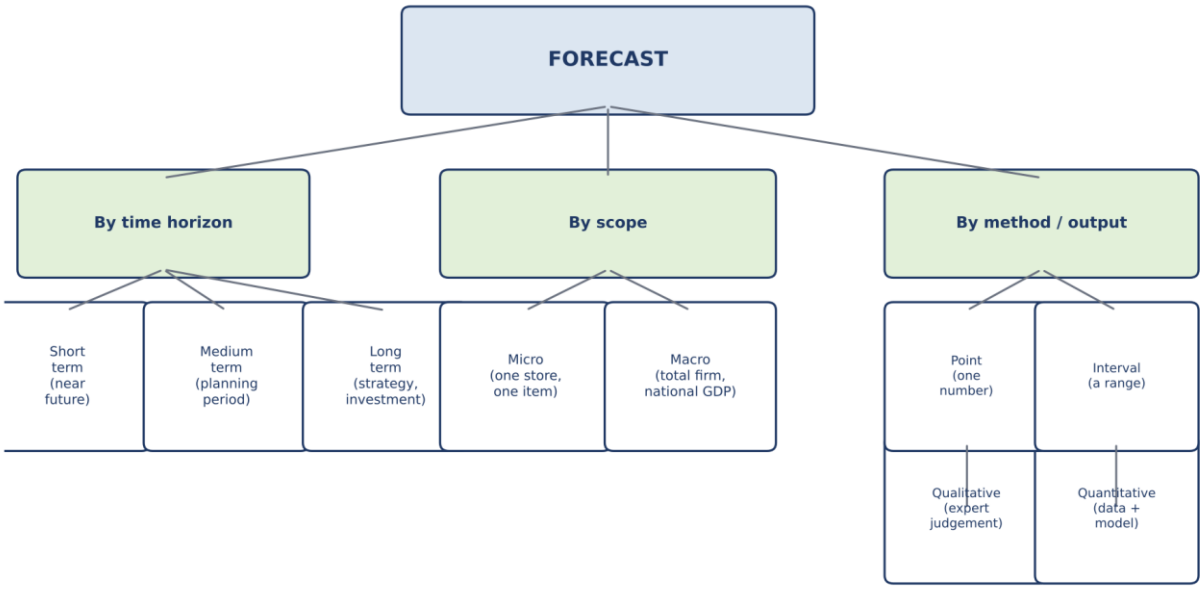


Figure 1. Forecasts can be classified by time horizon, by scope, and by method/output — a single forecast is usually described by picking one option from each branch.

Type	Simple meaning	Example
Short-term	Near future; supports day-to-day operations.	Next week's demand for a product.
Medium-term	Intermediate planning period.	Quarterly staffing or inventory plans.
Long-term	Distant future; supports strategy and investment.	Whether to build a new factory in 3 years.
Micro	Small, detail-level item or unit.	Sales at one store.
Macro	Large, aggregate measure.	National GDP or total company sales.
Qualitative	Uses expert judgement and opinions.	Forecasting demand for a brand-new product.
Quantitative	Uses numerical data and a calculation.	Forecasting next month's sales from past sales.

Type	Simple meaning	Example
Point forecast	One single best-estimate number.	Expected demand = 1,200 units.
Interval forecast	A range that expresses uncertainty.	Expected demand is between 1,100 and 1,300 units.

Important: "short/medium/long" have no universal fixed cut-off — the meaning depends on the industry and the decision being made.

✓ **3-MARK ANSWER — Q2. Distinguish between a point forecast and an interval forecast.**

1. A point forecast gives one single best-estimate number, e.g. "expected demand = 1,200 units."

2. An interval forecast gives a range that expresses the uncertainty around that estimate, e.g. "expected demand is between 1,100 and 1,300 units."

3. Interval forecasts are more informative for risk-based decisions because they show how confident the forecaster actually is.

1.3 Qualitative forecasting methods

Qualitative methods are used when numeric history is scarce or unreliable — for a brand-new product, for example — so the forecast leans on structured expert judgement instead.

Delphi method

A coordinator collects forecasts from a panel of experts independently, summarises the responses anonymously, and circulates the summary so experts can revise their view. This repeats over several rounds until opinions become reasonably stable.

> **IN PLAIN ENGLISH**
> Imagine asking five senior colleagues separately (so no one is influenced by a loud voice in the room), sharing the anonymous summary, and asking each to reconsider. After two or three rounds, their estimates usually converge.

- Why use it: gets informed judgement when data are scarce, and avoids a dominant personality skewing the result.
- Limit: it takes time and still depends on the quality of the experts' judgement.

Analogy

Estimate a new situation by comparing it with a genuinely similar past situation.

> **REAL-LIFE EXAMPLE**
> A company launching a new ice-cream flavour forecasts its first month's sales using the launch pattern of a similar flavour launched last year, adjusting for any real differences between the two launches.

Scenario forecasting

Instead of pretending there is only one future, the forecaster builds several plausible futures — for example, low-growth, expected, and high-growth demand — and states the assumption behind each one.

Judgemental adjustment

Start from a statistical forecast and adjust it using information the model could not have captured — such as a planned price rise or a one-off supply disruption. The size and reason for every adjustment should be recorded and later checked for whether it actually improved accuracy.

Important: never adjust a forecast just because the number "feels wrong" — only adjust for a specific, nameable reason.

> **✓ 3-MARK ANSWER — Q3. Explain the Delphi method and state one limitation.**
>
> **1.** Delphi collects forecasts from a panel of experts independently, so no one is influenced by the others.
>
> **2.** Responses are summarised anonymously and shared back; experts may revise their estimate, and the process repeats for a few rounds.
>
> **3.** It is useful when data are scarce, but it is time-consuming and the result still depends entirely on the quality of the experts chosen.

1.4 Components of a time series

A time series is simply a sequence of observations recorded over time — daily orders, monthly sales, quarterly revenue. The order matters: shuffling the same numbers randomly destroys the pattern.

Figure 2. Breaking a time series into its components

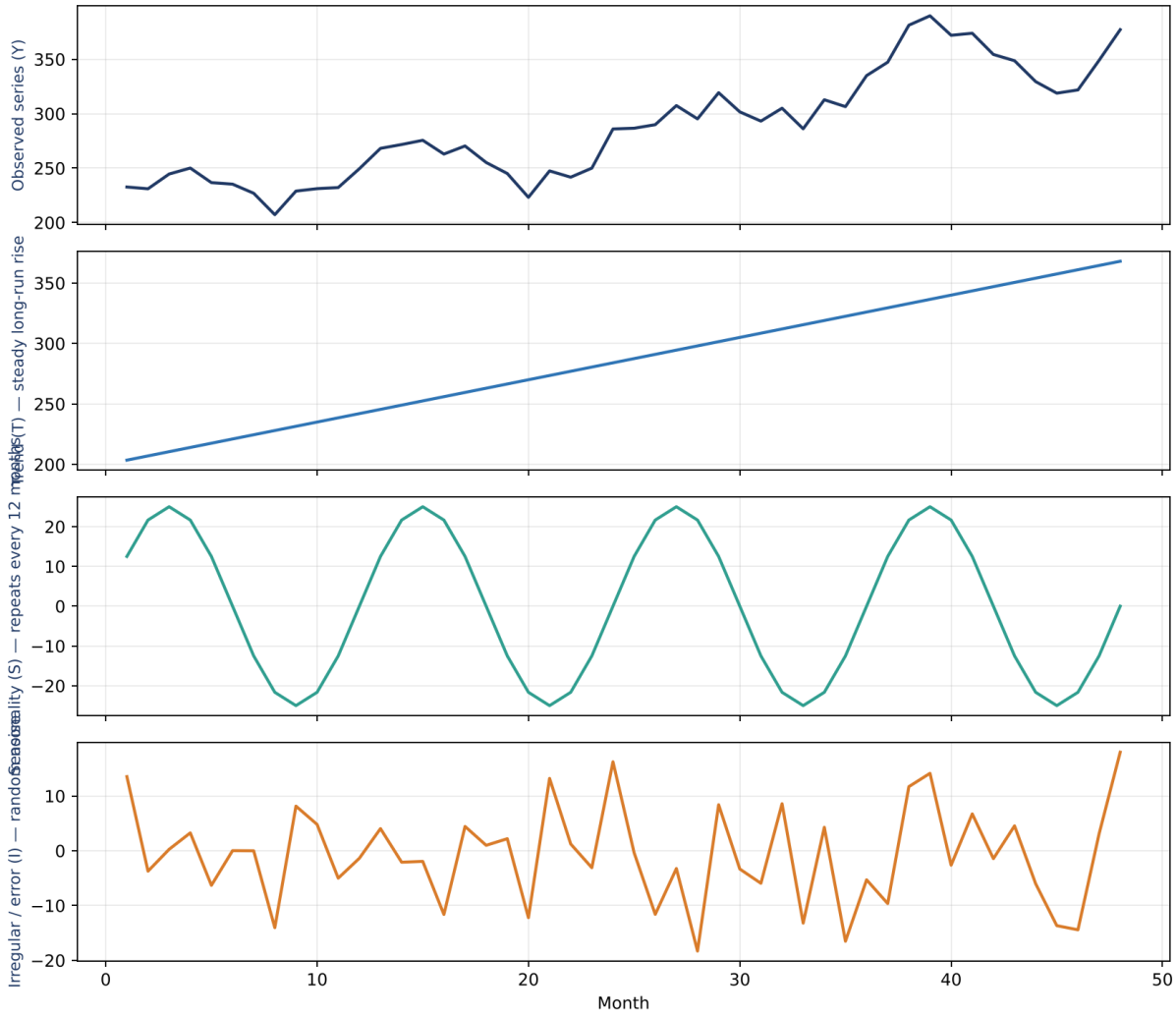


Figure 2. A real series (top) is usually the sum of a trend, a seasonal pattern, and random irregular noise. A cyclical wave can also be present over a longer horizon.

Component	Meaning	Real-life example
Trend (T)	Long-run upward or downward movement.	A city's population growing steadily each year.
Seasonality (S)	A pattern that repeats at a known, fixed calendar frequency.	Higher ice-cream sales every summer.
Cyclical (C)	Wave-like rises and falls around the trend; length is not fixed.	A multi-year boom-and-slowdown business cycle.
Irregular / error (I)	Unexpected variation left after the other patterns are removed.	A one-off strike or storm disrupting sales.

Two common ways to combine them: additive $Y = T + S + C + I$ (useful when seasonal swings stay roughly constant in size), and multiplicative $Y = T \times S \times C \times I$ (useful when the swings grow as the series level grows).

✓ 3-MARK ANSWER — Q4. Distinguish between seasonality and cyclical movement.

1. Seasonality repeats at a known, fixed calendar frequency (e.g. every December); cyclical movement has no fixed length.

2. Seasonality is usually tied to the calendar (holidays, weather); cycles are usually tied to broader economic or business forces.

3. Example: higher retail sales every December is seasonality; a multi-year boom-and-slowdown pattern is cyclical.

Stationary versus non-stationary

A series is stationary when its statistical behaviour — especially its mean (average level) and variance (spread) — stays broadly stable over time. A non-stationary series changes its level, spread, or dependence pattern.

REAL-LIFE EXAMPLE

Stationary: demand fluctuates around roughly 500 units a month, with a similar spread, month after month.

Non-stationary: sales rise from around 100 to around 1,000 over several years because of a sustained growth trend.

Why it matters: a method built for a stable level (like a simple moving average) can lag behind a real trend. Choosing the right method starts with recognising whether the series is stationary.

1.5 Autocorrelation (ACF) and the correlogram

Autocorrelation measures how strongly a series is related to its own past values. Instead of asking "are sales related to advertising?", it asks "are this month's sales related to last month's sales, two months ago, and so on?"

```
r_k  ≈  correlation between  Y_t  and  Y_(t-k)
```

Lag (k) means how many time steps apart the two values being compared are. With weekly data, lag 1 means one week apart; lag 4 means four weeks apart. The ACF value usually lies between -1 and +1: positive means values tend to move together, negative means a high value tends to be followed by a low one, and near zero means little linear relationship at that lag.

A correlogram is simply a bar chart of ACF values across lags — the standard picture used to read these relationships at a glance.

Figure 3. Correlogram of Sunrise Bakery weekly sales

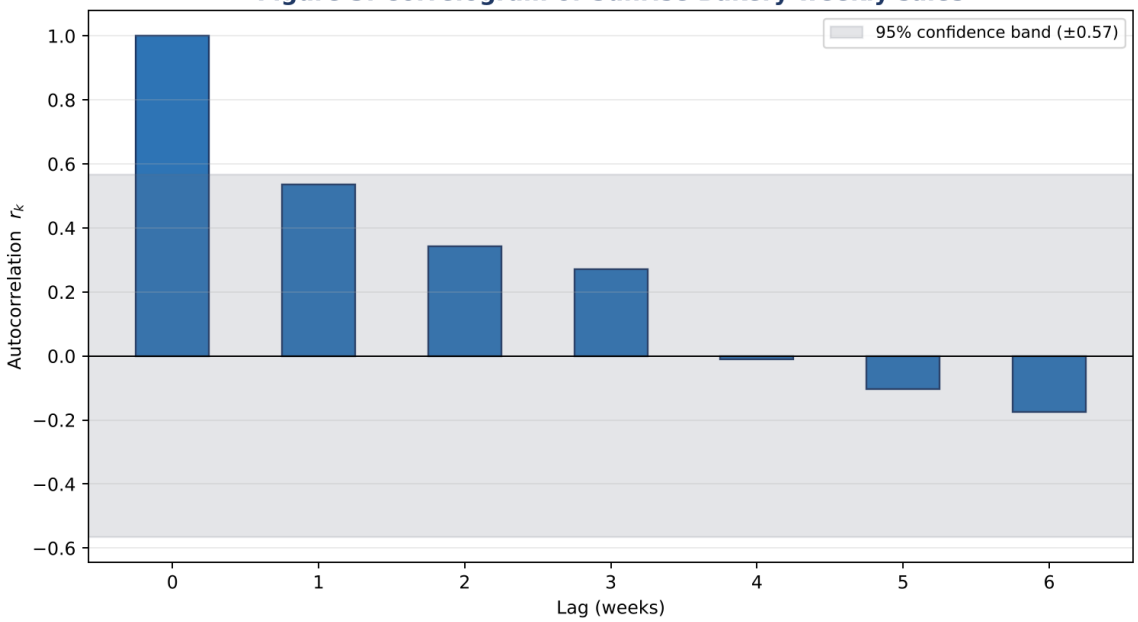


Figure 3. Correlogram of 12 weeks of Sunrise Bakery sales. The grey band is the approximate 95% confidence range — bars that cross it are larger than you'd expect from pure randomness.

ACF pattern	Possible meaning	What to check next
Large positive values that decay slowly	Persistence; may indicate trend or non-stationarity.	Plot the series and assess trend; don't conclude from ACF alone.
Spikes at seasonal lags (e.g. 12 for monthly data)	Possible annual seasonality.	Compare with the time plot and business calendar.
Values drop quickly near zero	Short-lived dependence; potentially a more stable series.	Check residuals and other diagnostics.
Most spikes inside the confidence band	No single lag is individually significant.	Several lags jointly may still matter — use Ljung-Box.

✓ 3-MARK ANSWER — Q5. What does the ACF/correlogram tell you about a series?

ACF pattern	Possible meaning	What to check next

1. The ACF measures how strongly a series is correlated with its own lagged (past) values, for lag 1, 2, 3, and so on.
2. A correlogram is a bar chart of these ACF values; bars that cross the confidence band suggest a relationship stronger than random chance.
3. A slowly decaying ACF warns of trend/non-stationarity; spikes at a fixed lag (e.g. 12 for monthly data) warn of seasonality — always confirm with the time plot, since the ACF alone is diagnostic evidence, not proof.

1.6 Forecast error metrics — MAD, MSE, MAPE, MPE

Once a forecast is made, it must be checked against what actually happened. Let Y_t be the actual value and $\hat{Y}_t$ be the forecast for the same period. The error (residual) is defined as:

$$e_t = Y_t - \hat{Y}_t$$

If actual = 120 and forecast = 110, the error is +10 (under-forecast by 10). If actual = 90 and forecast = 100, the error is -10 (over-forecast by 10). Always keep this sign convention consistent.

Tiny worked example (from the textbook)

Actuals: 100, 120, 80. Forecasts: 90, 130, 100.

Actual	Forecast	Error e	|e|	e^2	|e|/Actual × 100	e/Actual × 100
100	90	10	10	100	10.00%	10.00%
120	130	-10	10	100	8.33%	-8.33%
80	100	-20	20	400	25.00%	-25.00%

$$\text{MAD} = (10 + 10 + 20) / 3 = 13.33$$
$$\text{MSE} = (100 + 100 + 400) / 3 = 200$$
$$\text{MAPE} = (10.00\% + 8.33\% + 25.00\%) / 3 = 14.44\%$$
$$\text{MPE} = (10.00\% - 8.33\% - 25.00\%) / 3 = -7.78\%$$

Metric	Formula	Meaning	Watch out for
MAD	average of |error|	Average size of the miss, in original units.	Removes the sign, so it hides the direction of bias.
MSE	average of error²	Average squared miss; large errors are punished much more.	Units are squared and sensitive to outliers.
MAPE	average of |error/actual| × 100	Average absolute % error; easy to compare across scales.	Undefined/unstable when actual is zero or near zero.
MPE	average of (error/actual) × 100	Average signed % error; shows bias direction.	Positive and negative errors can cancel out.

✓ 3-MARK ANSWER — Q6. Differentiate between MAD, MSE, MAPE and MPE.

1. MAD is the average absolute error in the original units; it shows typical error size but not direction.
2. MSE squares each error before averaging, so it penalises large misses far more heavily than small ones.
3. MAPE expresses the average absolute error as a percentage of the actual value, making it comparable across different scales, while MPE keeps the sign and so reveals whether the forecast is systematically biased high or low.

1.7 Error analysis: is the model actually adequate?

A residual is the part of an observation the forecast missed. After fitting a reasonable model, residuals should ideally look like unpredictable noise — no pattern left for a smarter model to exploit. Four separate questions are usually asked:

- Are the errors small? → MAD / MSE / MAPE
- Are the errors unbiased on average? → mean error / MPE
- Are the errors independent / random? → residual ACF / Ljung-Box test
- Is the spread stable and the shape plausible? → residual plot / homoscedasticity / normality checks

Homoscedasticity (constant variance)

"Homo" means same; "scedasticity" relates to spread. Homoscedastic residuals have roughly similar variance over time. A funnel shape on a residual plot — small errors early, much larger errors later — suggests heteroscedasticity (changing variance), which makes prediction intervals less reliable even if the point forecast itself isn't biased.

Shapiro-Wilk test for normality

This test asks whether a sample of residuals is consistent with a normal (bell-shaped) distribution. It is applied to model residuals, not to the raw forecasts.

p-value	Decision	Correct wording
$p > 0.05$	Fail to reject H_0	Not enough evidence to say the residuals depart from normality.
$p \leq 0.05$	Reject H_0	There is evidence that the residuals are not normally distributed.

Caution: never say "p > 0.05 proves normality." A test can simply fail to detect a departure, especially in a small sample.

Ljung-Box test

The Ljung-Box test checks whether a group of autocorrelations up to a chosen lag is collectively different from zero — in other words, whether the residuals behave like independent noise.

$$Q = n(n+2) \sum [\, r_k^2 / (n-k) \,], \quad \text{for } k = 1,\ldots,m$$

p-value	Decision	Meaning
$p > 0.05$	Fail to reject H_0	No significant group autocorrelation detected through the chosen lag.
$p \leq 0.05$	Reject H_0	Evidence of remaining serial correlation — the model may have missed structure.

✓ 3-MARK ANSWER — Q7. Explain the Ljung-Box test and the Shapiro-Wilk test, and how they differ.

1. Ljung-Box tests whether a group of residual autocorrelations (up to a chosen lag) is jointly different from zero — i.e. whether the residuals behave like random noise.

2. Shapiro-Wilk tests whether the residuals are consistent with a normal distribution — a completely different question about the shape of the error distribution.

3. Passing one test does not mean the other is passed: residuals can be uncorrelated (pass Ljung-Box) yet still non-normal (fail Shapiro-Wilk), or vice-versa, so both are checked separately.

UNIT 2 — Stationary Models

⏱ WHY THIS UNIT HAS 6 LEARNING HOURS

Unit 2 moves from concepts into real computation: you now calculate forecasts by hand and in Excel for four different methods on the same data.

The extra hours go into mastering the mechanics — correctly shifting the averaging window, applying weights, chaining two averages together, and tuning a smoothing constant — not into harder theory.

2.1 Simple Moving Average (SMA)

A simple moving average forecasts the next period by averaging the most recent k actual observations, each with equal weight. As a new actual arrives, the oldest observation is dropped — hence "moving."

$$\text{Forecast for } t+1 = (Y_t + Y_{(t-1)} + \ldots + Y_{(t-k+1)}) / k$$

IN PLAIN ENGLISH

It's like judging how busy a cafe usually is by averaging its last 3 days of footfall, and dropping the oldest day each time a new one comes in — a rolling, always-up-to-date typical level.

Why use it: it smooths short-term random ups and downs and gives a simple baseline forecast when the series fluctuates around a fairly stable level. Limitation: because it averages in older, lower values along with newer ones, it can lag behind a real trend.

Worked example — 3-period SMA

Suppose the last three actual demands are 100, 110, and 120. The next forecast is:

$$(100 + 110 + 120) / 3 = 110 \text{ units}$$

If the next actual turns out to be 130, the following forecast drops 100 and uses 110, 120, 130: (110+120+130)/3 = 120.

✓ 3-MARK ANSWER — Q8. Explain Simple Moving Average and state one strength and one weakness.

1. SMA forecasts the next period as the equally-weighted average of the most recent k actual values.
2. Strength: it is simple to compute and smooths out short-term random fluctuation, giving a stable baseline forecast.
3. Weakness: because older and newer values are weighted equally, SMA lags behind the data whenever a real trend is present.

2.2 Weighted Moving Average (WMA)

WMA is like SMA, but the observations do not all receive equal importance. The forecaster chooses weights, usually giving the most recent observation the largest weight.

$$\text{Forecast} = w_1 Y_{t-1} + w_2 Y_{t-2} + \ldots + w_k Y_{t-k}$$
$$\text{Weights must add to 1:} \quad \Sigma w_i = 1$$

Worked example

Actuals, oldest to newest: 100, 110, 120. Give weights 1/6, 2/6, 3/6, so the newest value gets the greatest weight:

$$\text{Forecast} = (100 \times 1/6) + (110 \times 2/6) + (120 \times 3/6)$$
$$= 16.67 + 36.67 + 60 = 113.33$$

This reacts more to the recent rise than the SMA forecast of 110 did — WMA is more responsive because it deliberately leans on the newest data.

✓ 3-MARK ANSWER — Q9. How does WMA differ from SMA, and why might it forecast differently?

1. SMA gives every observation in the window equal weight; WMA assigns different (forecaster-chosen) weights that must sum to 1.

2. WMA typically puts the largest weight on the most recent observation, so it reacts faster to a recent change than SMA does.

3. On the same data, WMA and SMA can give different forecasts — e.g. 113.33 (WMA) versus 110.00 (SMA) — because WMA leans harder on the newest value.

2.3 Double Moving Average (DMA)

DMA is designed for a series with a roughly linear trend — something SMA and WMA handle poorly. First calculate a moving average of the original series (the first moving average, M_t). Then calculate a moving average of those moving averages (the second moving average, M'_t). The gap between the two estimates the trend that a single moving average tends to lag behind.

```
First MA:  M_t = average of the latest k actual values
Second MA: M'_t = average of the latest k first-MA values
                    a_t = 2M_t − M'_t
                b_t = [2/(k-1)](M_t − M'_t)
Forecast p periods ahead:  Ŷ_(t+p) = a_t + b_t·p
```

a_t is the estimated current level, b_t is the estimated trend per period, and p is how many periods ahead you are forecasting.

Worked example (k = 3)

Suppose the latest first moving average is M_t = 120 and the latest second moving average is M'_t = 110:

```
a_t = 2(120) − 110 = 130
b_t = [2/(3-1)](120-110) = 10
One-step forecast = 130 + 10(1) = 140
Two-step forecast = 130 + 10(2) = 150
```

The estimated level (130) can be ahead of the first MA (120) because the method is specifically correcting for the lag — that is not a mistake, it is the whole point of DMA.

✓ 3-MARK ANSWER — Q10. Why is Double Moving Average used instead of a single moving average?

1. A single moving average lags behind a trending series because it keeps averaging in older, lower (or higher) values.

2. DMA takes a moving average of the moving average itself; the gap between the two averages is used to estimate the trend (b_t) and correct the level (a_t).

3. The corrected forecast, a_t + b_t·p, can then project the trend forward for p periods ahead, unlike SMA which effectively assumes no trend.

2.4 Simple Exponential Smoothing (SES)

Exponential smoothing updates the forecast by blending the latest actual observation with the previous forecast. It is called "exponential" because the influence of older observations decays exponentially the further back in time they are.

$$\hat{Y}_{t+1} = \alpha Y_t + (1-\alpha)\hat{Y}_t, \quad \text{where } 0 \le \alpha \le 1$$

α (alpha) is the smoothing constant. The newest actual value receives weight α; the old forecast carries the remaining weight (1-α).

Worked example

Previous forecast = 100, latest actual = 120, α = 0.3:

$$\text{New forecast} = 0.3(120) + 0.7(100) = 36 + 70 = 106$$

The forecast moves from 100 toward the new actual of 120, but only part of the way. A larger alpha would react more strongly to the new data.

Alpha value	Effect
Closer to 1	More weight on the latest actual; reacts quickly, but can chase noise.
Closer to 0	More weight on the previous forecast; smoother, but slower to adapt.

SES suits a series without a strong trend or seasonality. If a strong trend is present, SES forecasts will lag — a trend method such as Holt's method or DMA is then more appropriate.

✓ 3-MARK ANSWER — Q11. Explain the smoothing constant alpha in SES and how it is chosen.

1. Alpha controls how much weight the newest actual observation gets versus the previous forecast, with $0 \le \alpha \le 1$.

2. A larger alpha reacts faster to recent changes but can follow random noise; a smaller alpha is smoother but slower to adapt.

3. In practice alpha is often chosen by minimising the mean squared error (MSE) of past forecasts, for example using Excel Solver, rather than guessed arbitrarily.

Full Worked Case Study — Sunrise Bakery

Sunrise Bakery records its weekly sales of a popular sourdough loaf. The owner wants a week-13 forecast and wants to understand which method would have served her best over the last 12 weeks. All numbers below are genuinely computed from the 12 actual values — nothing is invented.

Week	1	2	3	4	5	6	7	8	9	10	11	12
Loaves sold	210	225	198	240	255	230	265	270	250	280	295	275

Step 1. Check the series for trend using the ACF

The correlogram (Figure 3, shown earlier) shows positive autocorrelation at lag 1 (0.535) that decays over the next couple of lags — a mild sign of persistence/trend, consistent with sales visibly climbing from around 210 in week 1 to 275-295 by the final weeks. This tells us SMA and WMA (level methods) may lag, and DMA (a trend method) is worth comparing.

Step 2. Simple Moving Average (k = 3)

Forecast for week	Uses weeks	Calculation	Forecast
4	1-3	(210+225+198)/3	211.00
5	2-4	(225+198+240)/3	221.00
6	3-5	(198+240+255)/3	231.00
7	4-6	(240+255+230)/3	241.67
8	5-7	(255+230+265)/3	250.00
9	6-8	(230+265+270)/3	255.00
10	7-9	(265+270+250)/3	261.67
11	8-10	(270+250+280)/3	266.67
12	9-11	(250+280+295)/3	275.00
13 (next)	10-12	(280+295+275)/3	283.33

Error metrics over weeks 4-12 (comparing each forecast to that week's actual): MAD = 17.67, MSE = 456.26, RMSE = 21.36, MAPE = 6.69%, MPE = +6.15%. The positive MPE confirms SMA is systematically under-forecasting — expected, since the series is trending up and SMA lags.

Step 3. Weighted Moving Average (weights 1/6, 2/6, 3/6)

Forecast for week	Calculation	Forecast
4	210(1/6)+225(2/6)+198(3/6)	209.00
7	240(1/6)+255(2/6)+230(3/6)	240.00
10	265(1/6)+270(2/6)+250(3/6)	259.17
13 (next)	280(1/6)+295(2/6)+275(3/6)	282.50

Error metrics over weeks 4-12: MAD = 20.33, MSE = 484.68, RMSE = 22.02, MAPE = 7.77%, MPE = +5.11%. On this dataset, WMA's heavier recent weighting did not reduce MAD/MSE versus SMA — a reminder that "more responsive" is not automatically "more accurate"; it depends on how noisy the latest observation is.

Figure 4. Actual sales vs SMA and WMA forecasts — Sunrise Bakery

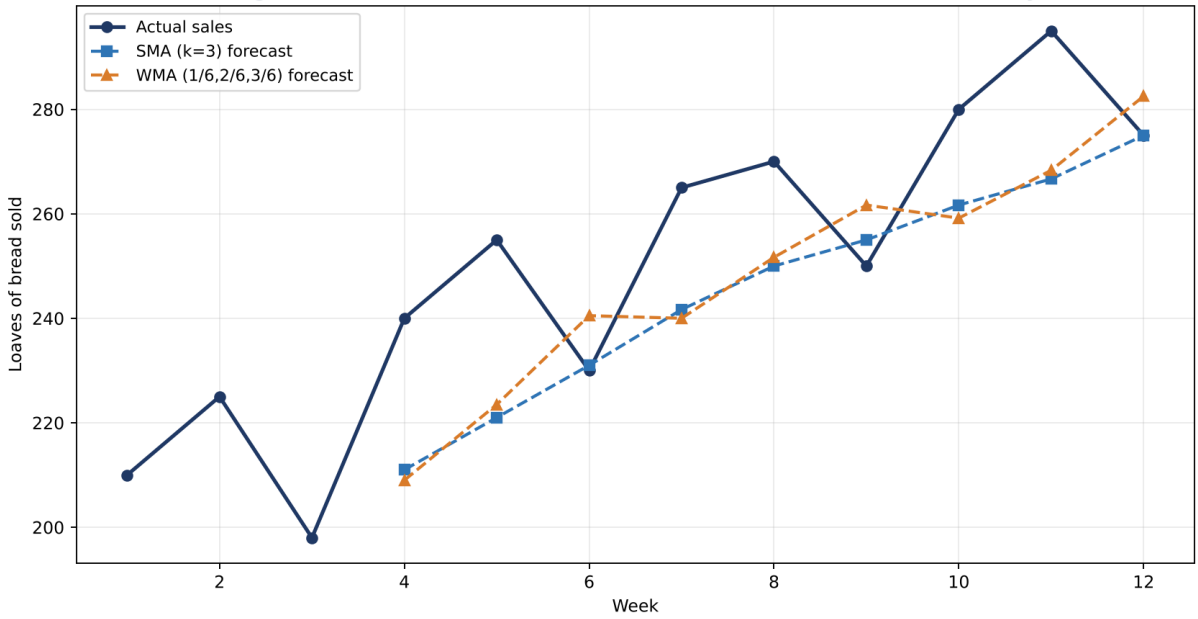


Figure 4. SMA and WMA forecasts both visibly trail behind the rising actual sales line — the classic sign that a level-only method is being used on a trending series.

Step 4. Double Moving Average (k = 3)

Using the first moving averages already calculated for SMA as M_t, and taking a 3-period moving average of those to get M'_t:

Period	First MA (M_t)	Second MA (M'_t)
5	231.00	221.00
8	255.00	248.89
11	275.00	267.78
12	283.33	275.00

At week 12 (the latest point): M_t = 283.33, M'_t = 275.00.

```
a_t = 2(283.33) - 275.00 = 291.67
b_t = [2/(3-1)](283.33-275.00) = 8.33
Week 13 forecast (p=1): 291.67 + 8.33(1) = 300.00
Week 14 forecast (p=2): 291.67 + 8.33(2) = 308.33
```

Figure 5. Double Moving Average projects the trend forward

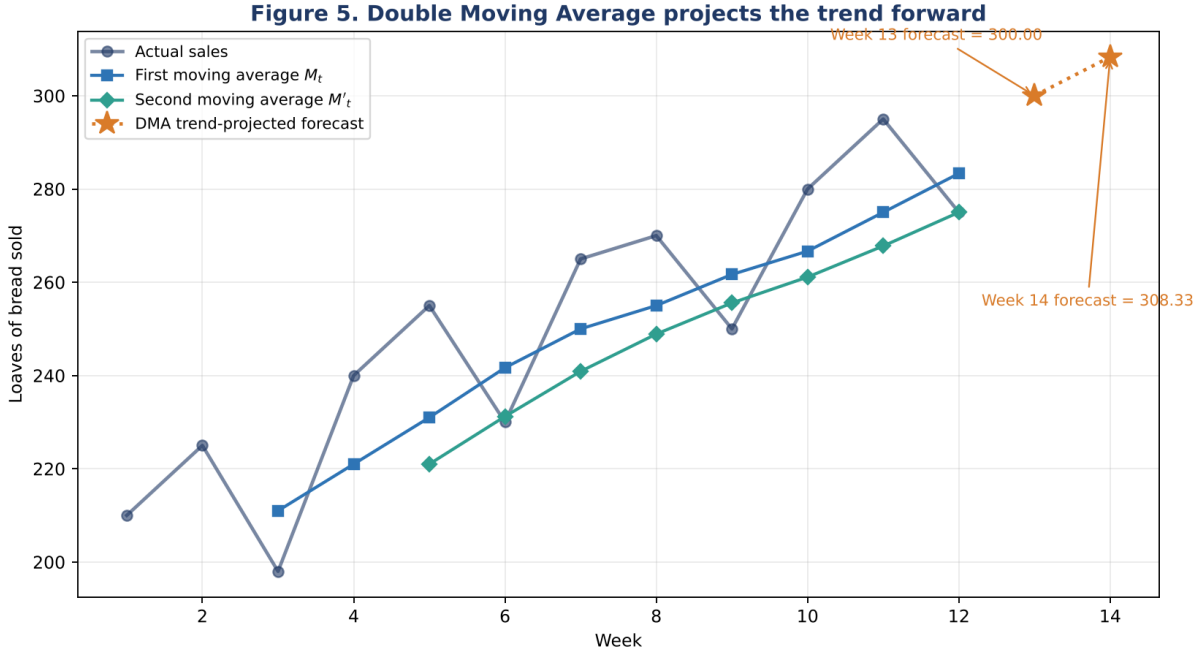


Figure 5. DMA tracks the first and second moving averages and projects the gap between them forward as a trend — giving week 13 and 14 forecasts of 300.00 and 308.33.

DMA's week-13 forecast of 300.00 is noticeably higher than SMA's 283.33 and WMA's 282.50 — because DMA is the only one of the three that explicitly accounts for the upward trend instead of just averaging recent levels.

Step 5. Simple Exponential Smoothing (α = 0.3, and a fitted α)

Starting with the week-1 actual (210) as the initial forecast, and α = 0.3:

Week	Actual	SES forecast (α=0.3)
2	225	210.00
5	255	218.68
8	270	240.29
11	295	258.61
13 (next)	—	271.17

Error metrics (weeks 2-12, α=0.3): MAD = 21.54, MSE = 650.91, RMSE = 25.51, MAPE = 8.36%, MPE = +6.85%. This is the weakest of the four methods on this dataset — exactly as the theory predicts, since SES is a level method and this series has a clear trend.

Searching across candidate alpha values for the one that minimises MSE on this data gives a best-fit $\alpha \approx 0.59$ (MSE = 495.35), which reacts faster to recent weeks and performs better than $\alpha = 0.3$ — but is still not as good as DMA, because no SES value of alpha can fully correct for a trend; it can only react faster to it.

Figure 6. Effect of the smoothing constant alpha on SES forecasts

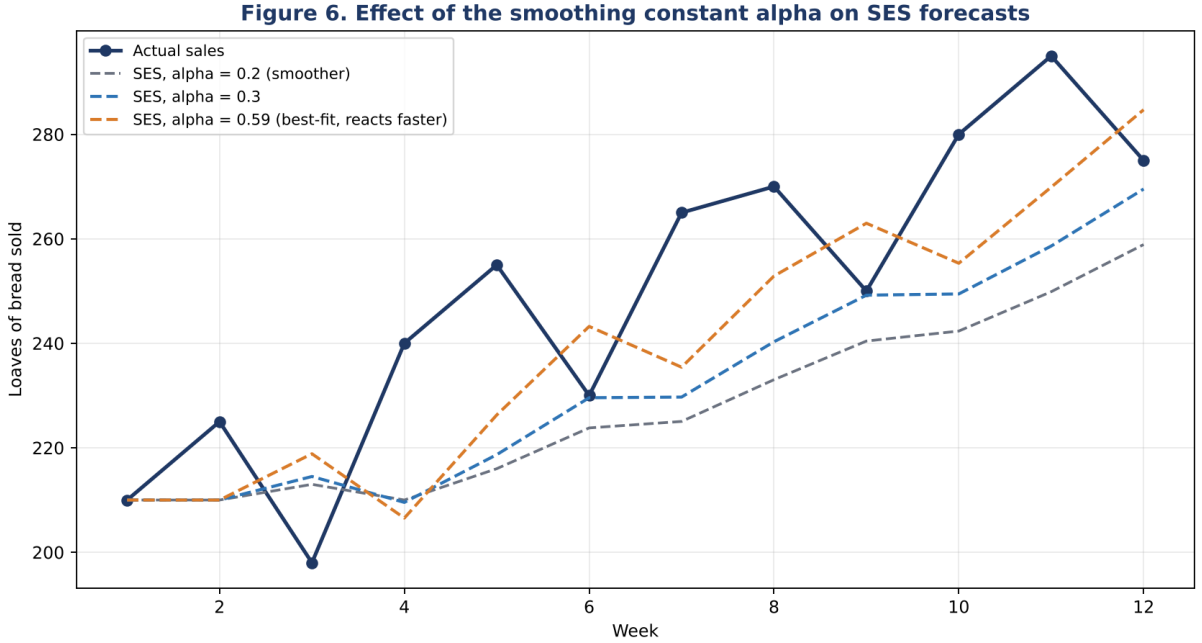


Figure 6. A larger alpha (0.59) tracks the rising actual sales more closely than a smaller alpha (0.2 or 0.3), but SES — whatever the alpha — is still structurally a level method, not a trend method.

Step 6. Comparing all four methods

Method	MAD	MSE	MAPE	MPE	Week-13 forecast
SMA (k=3)	17.67	456.26	6.69%	+6.15%	283.33
WMA (1/6,2/6,3/6)	20.33	484.68	7.77%	+5.11%	282.50
SES (α=0.3)	21.54	650.91	8.36%	+6.85%	271.17
SES (α=0.59, fitted)	—	495.35	—	—	~277
DMA (k=3)	—	—	—	—	300.00

Conclusion: on raw error metrics over the fitting window, SMA happens to look best among the level methods, but every level method (SMA, WMA, SES) shows a consistent positive MPE — a tell-tale sign of under-forecasting a trending series. DMA is the only method structurally built to correct that bias, which is exactly why it is the recommended method once a trend has been confirmed via the ACF and the time plot.

✓ 3-MARK ANSWER — Q12. Based on the Sunrise Bakery case study, which forecasting method would you recommend and why?

1. The correlogram and the rising weekly sales both indicate a trend, which level methods (SMA, WMA, SES) are not designed to track.

2. All three level methods show a positive MPE on this data, meaning they systematically under-forecast as sales keep climbing.

3. Double Moving Average is recommended because it explicitly estimates and projects the trend (giving a_t and b_t), producing a higher and more trend-consistent week-13 forecast of 300.00 compared with 283.33-282.50 from the level methods.

Common Exam and Excel Mistakes

- Using forecast − actual when the course defines error as actual − forecast.
- Taking the average of signed errors and calling it MAD (MAD must use absolute values).
- Forgetting to square errors for MSE, or forgetting the square root for RMSE.
- Forgetting to ×100 when reporting MAPE/MPE as percentages.
- Using MAPE when an actual value is zero (undefined).
- Comparing different periods/rows instead of the same valid window.
- Using the current period's actual value inside its own forecast ("look-ahead" leakage).
- Giving WMA's largest weight to the oldest value by mistake instead of the newest.
- Using simple exponential smoothing on a strong trend without checking whether a trend method is needed instead.
- Claiming $p > 0.05$ "proves" normality or "proves" no autocorrelation — it only means not enough evidence to reject.
- Treating randomness, constant variance (homoscedasticity), and normality as if they were the same condition — they are three separate checks.
- Interpreting a single ACF spike without checking the lag number, the data frequency, and the overall pattern.

Rapid Revision Sheet

Question	Remember
Forecast error?	Actual − forecast.
MAD?	Average absolute error; original units; no direction.
MSE?	Average squared error; punishes large misses.
MAPE?	Average absolute % error; breaks down when actual = 0.
MPE?	Average signed % error; shows bias direction; can cancel out.
Stationary?	Stable statistical behaviour, especially mean and variance.
ACF?	Correlation of a series with its own lagged values.
Slowly decaying ACF?	May indicate persistence / trend / non-stationarity.
Seasonal ACF spikes?	Spikes at regular seasonal lags (e.g. 12 for monthly data).
Ljung-Box $p \leq .05$?	Reject “no autocorrelation”; residual structure remains.
Shapiro-Wilk $p \leq .05$?	Reject normality; evidence residuals are not normal.
Homoscedasticity?	Roughly constant error variance over time.
SMA?	Equal weights across the latest k actuals.
WMA?	Unequal, chosen weights; must sum to 1.

Question	Remember
DMA?	Two chained moving averages estimate level AND trend.
SES?	Next forecast = α × latest actual + (1-α) × previous forecast.

BEFORE YOU SUBMIT AN ANSWER

☐ All metrics use the same, intended residual column and the same valid rows.

☐ Alpha lies between 0 and 1; WMA weights sum to 1.

☐ Numbers are interpreted in words, with correct units/percentages.

☐ When asked which model is best, consider accuracy AND residual diagnostics — not just one metric.